

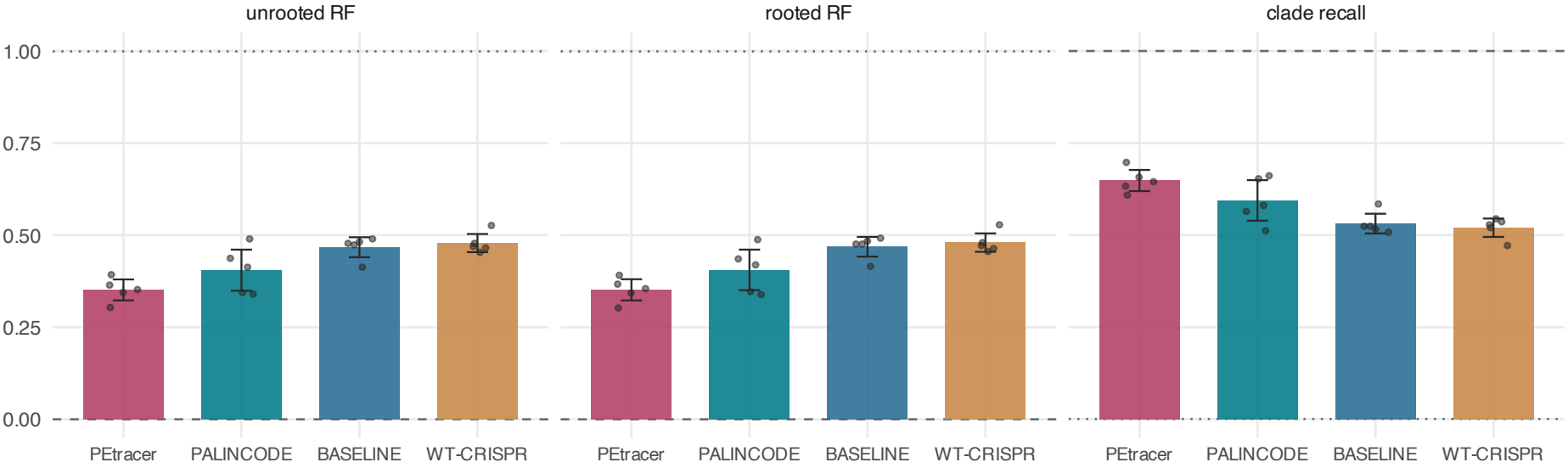
Lineage recorder comparison at matched capacity

Information matched at ~120-155 bits: BASELINE 2x50, WT-CRISPR 13x10, PEtracer 14x3 (9-state), PALINCODE 30x2

A

Topology recovery

dashed = perfect recorder, dotted = random tree (RF lower is better, recall higher)



B

Rooted ancestry and distance geometry

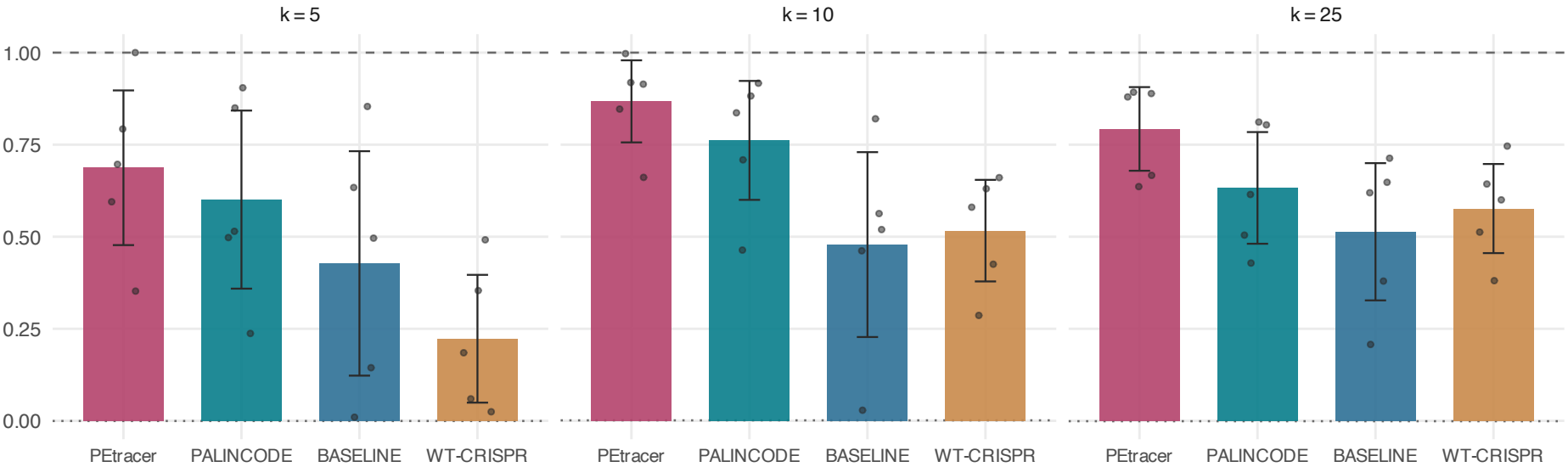
chance is 0.33 for triplets, 0.50 for ordering, 0.00 for rho (higher is better)



C

Clone assignment (adjusted Rand index)

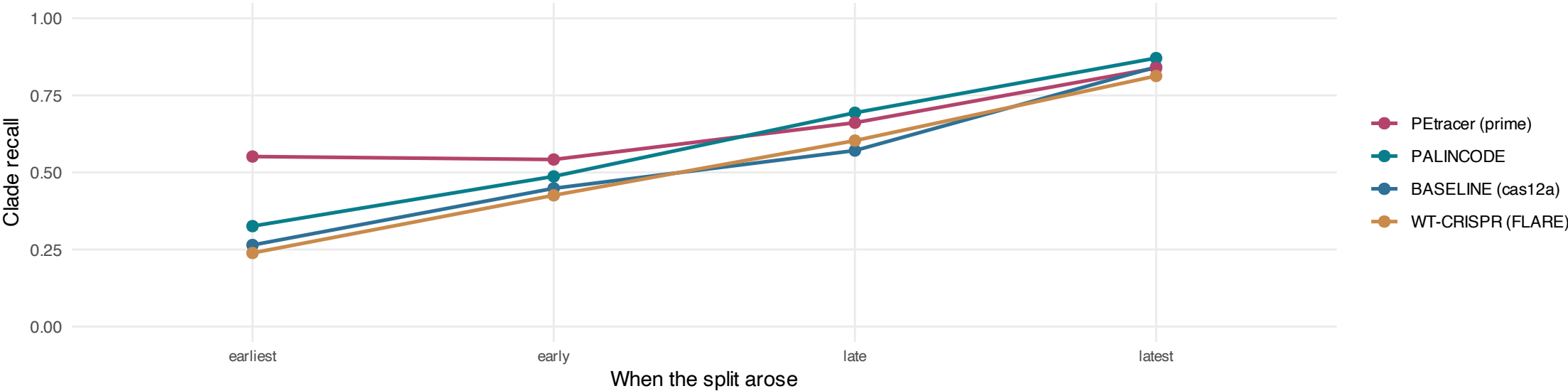
both trees cut into the same number of groups, so this scores assignment (higher is better)



D

Which splits are recovered: ancient or recent

True clades binned by when they arose (the truth tree is ultrametric, so node time is literally split time).
A steep slope means a recorder resolves recent structure but loses deep structure



Bars are means over 5 seeds; points are individual seeds. The perfect-recorder reference is neighbour joining run on the true patristic distances, and it recovers the tree exactly on every topology metric -- so NJ imposes no ceiling here and all observed error is the recorder's, not the method's. Clade precision equals recall by construction: both trees are fully resolved and binary, so they contain the same number of clades and the two ratios share a denominator.